

Vortex-induced vibration of a functionally graded metamaterial plate attached to a cylinder in laminar flow

Bill Murari, Shaoyu Zhao, Yihe Zhang, Jie Yang^{*}

School of Engineering, RMIT University, PO Box 71, Bundoora, VIC 3083 Australia

ARTICLE INFO

Keywords:

Vortex-induced vibration
Functionally graded plate
Graphene origami
Fluid
Auxetic material
Negative Poisson's ratio
Kármán vortex street

ABSTRACT

Understanding the interaction between fluid and structures under various fluid conditions is critical for engineering applications, with vortex-induced vibration (VIV) being a significant area of research. This study presents a comprehensive numerical investigation on the VIV behaviors of functionally graded (FG) graphene origami (GOri)-enabled auxetic metamaterial (GOEAM) splitter plates attached to a circular cylinder by employing the finite element analysis. The splitter plate is composed of multiple GOEAM layers with the material properties possessing a gradient variation along the thickness direction. The local mesh quality is monitored and improved by Yeoh smoothing. The present work is focused on the effect of the auxetic properties of GOEAM on plate vibrations, the influence of plate length on wake patterns, and the effect of fluid loads on the cylinder-plate body and vortex pattern. The study reveals that: (1) the plate structure with H_{GRX} - W_{GRX} metamaterial distribution pattern with high elastic moduli in surface layers exhibits low deflection amplitudes, (2) shorter plates intensify vortices while longer plates reduce vortex prevalence, (3) the natural frequencies of both FG-GOEAM and pure metallic plates are found to be well above the vortex shedding frequency and as such, the fluid loads have a negligible effect on the cylinder-plate body and vortex patterns. These insights offer an in-depth understanding of fluid-structure interactions and valuable knowledge for the design and optimisation of graphene based metamaterial structures in engineering applications.

1. Introduction

Metamaterial structures possess unique physical properties that are not available in nature and have been attracting huge attention from both research and industry communities. In particular, mechanical metamaterials such as auxetic materials that exhibit negative Poisson's ratio (NPR) during deformation have demonstrated great promise for versatile functional and/or structural applications [1–3]. Graphene is a natural material with exceptional mechanical strength and stiffness. Introducing hydrogen atoms onto 2D graphene sheets through chemical modification process leads to the formation of creases that enable the resulting 3D graphene origami (GOri) to exhibit excellent auxetic properties [4]. GOri offers unique benefits compared to other materials due to its ability to be folded into intricate patterns and structures, allowing for precise control over mechanical properties such as NPR and Young's modulus. This flexibility in design enables tailored auxetic behaviour and improved mechanical performance in metallic metamaterials. In addition, the GOri reinforcement allows for exceptional dynamic response as a result of its inherent energy absorption

capabilities. Meng, et al. [5] demonstrated the hydrogenation-assisted self-assembly of nanoscale graphene Miura origami, which exhibits a negative Poisson's ratio and can be accurately predicted using a continuum model, providing insights into the design of three-dimensional nanoarchitectures from two-dimensional nanomaterials. Wei, et al. [6] discussed how to manipulate the shape and properties of graphene by creating creased marks on its surface, which can guide the folding process and allow for the construction of graphene-based nanodevices. Through molecular dynamics simulations, graphene metamaterials obtained by hydrogenation can exhibit a wide range of coefficients of thermal expansion [7]. Cai, et al. [8] discussed the significance of graphene with tuneable thermo-mechanical properties, specifically focusing on non-porous hydrogenated graphene origami metamaterials and their unique combination of tuneable high strength, enhanced stretchability, and thermal conductivity. Despite the above developments, maintaining its exceptional material strength and auxetic properties remains very challenging. Zhao, et al. [9] presented a methodology of introducing GOri into a metallic composite, creating a novel graphene origami-enabled auxetic metallic metamaterial (GOEAM).

^{*} Corresponding author.

E-mail address: j.yang@rmit.edu.au (J. Yang).

<https://doi.org/10.1016/j.tws.2024.111790>

Received 21 February 2024; Received in revised form 4 March 2024; Accepted 7 March 2024

Available online 9 March 2024

0263-8231/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

This approach involves incorporating a GORI structure with readily adjustable NPR characteristics into a copper matrix, aiming to improve the mechanical properties of the composite material. While these advancements signify notable strides, the evolution of metamaterial composite structures with both high-performance and customisable structural features is still in its nascent phase.

Functionally graded materials (FGMs) are a class of advanced structures whereby the material properties within the composite layers change smoothly and continuously depending on their spatial coordinates. Saleh, et al. [10] documented extensive applications of FGM structures, spanning across aerospace [11–14], automotive [15] and biomedical [16–19] sectors. Incorporating GOEAM metamaterials as reinforcements in a functionally graded setup has been receiving research attention due to the exceptional mechanical characteristics they exhibit. Lv, et al. [20] explores how FG-GOEAM plates behave under high-temperature compression and thermal loading. Chen, et al. [21] explores the nonlinear free vibration behaviour of a microplate reinforced by GOEAM functionally graded properties. By varying the graphene origami content across the thickness of the microplate, modifications to its auxetic property and other material characteristics can be made. Mathematical modelling, genetic programming, and numerical methods are used to analyse the nonlinear frequency and deflection of the microplate, showing potential for adjusting its vibration characteristics through manipulation of the GORI content and distribution [21]. The recent developments in FG-GOEAM structures demonstrate the enormous potential of their applications.

Understanding the fluid-structure interaction (FSI) under various fluid conditions is of paramount importance for engineering applications. Various studies have investigated the influence of static and dynamic fluids on functionally graded structures. Studies on the effect of FSI on the mechanical characteristics of FG-GOEAM beams, considering both linear and nonlinear aspects, have been conducted utilising the added mass method [22,23]. It was found that immersing beams in fluid amplifies their inertia characteristics and elevates the hydrodynamic pressure exerted on the structures. This effect dampens the natural frequency of the beams and prevents structural deformation by increasing their critical failure loads. Su, et al. [24] proposed a novel method for studying the behaviour of fluid-filled cylindrical shells made of FGMs. Pham, et al. [25] introduced a new method called isogeometric analysis for studying the vibrations of FGM rectangular plates in a fluid. The method is tested, and the effects of various factors on the vibrations of the plates are investigated. Dynamic fluid conditions such as vortex-induced vibration (VIV) have also been receiving research attention [26,27]. VIV occurs when fluid flow induces vibrations in structures, which can lead to fatigue damage and structural failure. Understanding and mitigating VIV is crucial in various engineering fields, including offshore structures (such as oil platforms), marine engineering (ship hulls), and civil infrastructure (bridges). However, most VIV studies have been limited to structures with uniform and conventional material properties. Consequently, the effects of the fluid flow on structures with graded and metamaterial properties are yet to be understood.

The plate-cylinder system represents a simplified model of a real-world scenario where flexible structures are subjected to fluid flow such as offshore structures, heat exchangers, and energy harvesting devices. According to Sun, et al. [28], the preferred option for obstruction is the circular cylinder, given its well-established comprehension of the vortex shedding process and Kármán vortex streets. Zhou, et al. [29] found that the shedding vortices in a Kármán vortex street can be harnessed to generate energy through VIV. Introducing a flexible plate into the downstream flow of the circular cylinder has the potential to induce large vibrational amplitudes through the periodic impingement of shed vortices. Dey [30] studied the effect of dual splitter plates on the wake topology around a square cylinder. According to a recent review on the control of VIV of circular cylinders, passive devices such as splitter plates, control rods, helical strakes, porous layers, and rough

surfaces have been found to effectively suppress VIV. Additionally, active flow control devices like jet flow and rotating control rods can achieve nearly full VIV suppression [31]. Chen, et al. [32] conducted a numerical investigation on the VIV of an elastically mounted circular cylinder with multiple control rods at low Reynolds number (Re), finding that the spacing ratio significantly influenced the kinematic response of the system and that using small control rods reduced vibration amplitude. Most studies investigating the FSI effect emphasise ideal fluid conditions such as steady and inviscid fluids, yet in order to improve the applicability of the FGM structures, their behaviour in dynamic and viscous fluid conditions should be well understood. Investigations into the fluid effect on these structures are still in their infancy [33,34].

To fill the aforementioned gaps, the effects of VIVs on FG-GOEAM splitter plates are comprehensively studied. In the following research, the fluid-domain setup described by Turek and Hron [35] is adopted, whereby, the VIV of a flexible splitter plate attached to a circular cylinder is studied using a finite element analysis and partial differential equation solver. This benchmark focuses on examining different coupling mechanisms, discretisation schemes, and the robustness of integrated solver components. It is based on successful configurations for incompressible laminar flow and involves an elastic object submerged in a channel flow. This setup results in self-induced oscillations of the structure, allowing for time-dependent analysis of characteristic physical quantities. The assembly is situated in a laminar flow with $Re = 200$. Kaneko, et al. [36] stated that the Strouhal number (St), a critical dimensionless parameter governing the shedding of vortices, for circular bodies lies within the range of $0.2 < St < 0.3$. The effects of metamaterial properties, plate length, and fluid loads on the plate vibrations, wake pattern of the flow, and cylinder-plate body and vortex pattern in the flow are explored, respectively. The method presented in the paper offers a distinctive advantage in handling problems related to functionally graded structures, particularly under vortex-induced vibration conditions. By focusing on these specific types of materials and their interactions with fluid flow, the study provides valuable insights that contribute to advancing our understanding of how such structures behave in engineering applications. The use of finite element analysis and partial differential equation solver tailored for functionally graded metamaterials sets this research apart by offering specialised tools to investigate these complex systems effectively.

2. Model

2.1. Structure-fluid system

Consider a two-dimensional computational domain as shown in Fig. 1. A circular cylinder of diameter D is placed within the domain. The aforementioned two-dimensional problem is numerically computed using a finite element-based solver. The distance between the centre of the circular cylinder and the inlet boundary of the domain is $2D$. The downstream distance of the domain from the centre of the circular

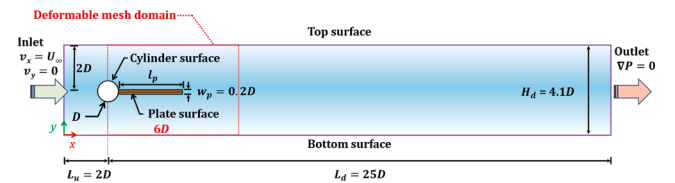


Fig. 1. Schematic depiction of the plate-fluid domain along with the boundary conditions. The size of the numerical domain is defined by the height, H_d , upstream length, L_u , and downstream length, L_d . The geometry of the splitter plate is defined by the length, L_p , width, w_p , v_x is the horizontal fluid velocity and v_y is the vertical fluid velocity.

cylinder is $23D$. The distance separating the top and bottom surfaces is $4.1D$, resulting in a blockage ratio ($D/4.1D$) of 24 %. The free-slip walls on the boundaries of the domain ensure that the fluid flow remains unperturbed by the top and bottom surfaces. A cantilever FG-GOEAM plate, with a plate length of $l_p = 3.5D$ and a plate thickness of $w_p = 0.2D$, is positioned downstream of the circular cylinder. The following are the boundary conditions imposed on the fluid-plate domain and are represented in non-dimensional form:

- 1) At the inlet ($X = 0, 0 \leq Y \leq H_d$), velocity inlet condition is placed, $U = 1$ and $V = 0$.
- 2) At the outlet ($X = L = L_u + L_d, 0 \leq Y \leq H_d$), outflow condition is imposed, $\partial U / \partial X = \partial V / \partial X = 0, P = 0$.
- 3) At the top and bottom surfaces of the domain ($Y = H_d, 0 \leq X \leq L = L_u + L_d$ and $Y = 0, 0 \leq X \leq L$), respectively, the free-slip condition is applied, $\partial U / \partial Y = 0$ and $V = 0$.
- 4) At the cylinder and the splitter plate surfaces, a no-slip condition is applied, $U = 0$ and $V = 0$.

U and V are the dimensionless velocity components along the longitudinal and transverse directions, respectively. P is the dimensionless pressure.

2.2. Metamaterial properties

The investigation focuses on FG-GOEAM plates, comprising a metallic matrix and GOr nanofillers. The graphene content exhibits a graded variation along the thickness direction, with the assumption that each layer of the matrix is isotropic. The formula provided by Lv, et al. [20] provides the volume fractions $V_{GR}(k)$ for the k^{th} layer ($k = 1, 2, 3, \dots, N_L$) under various distribution patterns, with N_L representing the total number of plate layers.

$$W_{GR} - U : V_{GR}(k) = V_{GR}^* \quad (1a)$$

$$W_{GR} - X : V_{GR}(k) = 2V_{GR}^* |2k - N_L - 1| / N_L \quad (1b)$$

$$W_{GR} - O : V_{GR}(k) = 2V_{GR}^* (1 - |2k - N_L - 1| / N_L) \quad (1c)$$

Here, V_{GR} stands for the volume fraction of GOr reinforcement in the k^{th} layer, while N_L indicates the total count of even layers in the FG-GOEAM plate. The overall graphene volume fraction, represented as V_{GR}^* , is determined from the weight fraction W_{GR} of graphene using the following conversion equation;

$$V_{GR}^* = \frac{1}{\left(1 + \frac{(1 - W_{GR})(\rho_{GR} / \rho_{metal})}{W_{GR}}\right)} \quad (2)$$

with ρ_{GR} and ρ_{metal} representing the mass densities of the graphene and metallic matrix, respectively.

The mathematical representations of the distribution patterns for the metamaterial are provided as algebraic expressions.

$$H_{GR} - U : H_{GR}(k) = H_{GR}^* \quad (3a)$$

$$H_{GR} - X : H_{GR}(k) = H_{GR}^* \cos\left(\frac{\pi x}{w_p}\right) \quad (3b)$$

$$H_{GR} - O : H_{GR}(k) = H_{GR}^* \left(1 - \cos\left(\frac{\pi x}{w_p}\right)\right) \quad (3c)$$

The above equations offer a depiction of the H coverage within the k^{th} layer of the FG-GOEAM plate, H_{GR}^* represents the uniform GOr folding degree (0 % - 100 %) and w_p is the plate thickness. The equations for the Young's modulus (E_c), density (ρ_c) and Poisson's ratio (ν_c) of the metamaterial are determined by the genetic programming (GP)-assisted

micromechanics models provided by Zhao, et al. [4].

$$E_c = \frac{1 + \xi \eta V_{GR}}{1 - \eta V_{GR}} E_{Cu} \times f_E(H_{GR}, V_{GR}, T) \quad (4a)$$

$$\nu_c = (\nu_{GR} V_{GR} + \nu_{Cu} V_{Cu}) \times f_\nu(H_{GR}, V_{GR}, T) \quad (4c)$$

$$\rho_c = (\rho_{GR} V_{GR} + \rho_{Cu} V_{Cu}) \times f_\rho(V_{GR}, T) \quad (4d)$$

where η and ξ are the material and size coefficients, respectively.

$$\eta = \frac{\left(\left(\frac{E_{GR}}{E_{Cu}}\right) - 1\right)}{\left(\frac{E_{GR}}{E_{Cu}}\right) + \xi} \quad (5)$$

$$\xi = 2 \left(\frac{l_{GR}}{t_{GR}}\right) \quad (6)$$

The length and thickness of graphene are l_{GR} and t_{GR} , respectively.

$$f_E(H_{GR}, V_{GR}, T) = 1.11 - 1.22V_{GR} - 0.134\left(\frac{T}{T_0}\right) + 0.559V_{GR}\left(\frac{T}{T_0}\right) - 5.5H_{GR}V_{GR} + 38H_{GR}V_{GR}^2 - 20.6H_{GR}^2V_{GR}^2 \quad (7a)$$

$$f_\nu(H_{GR}, V_{GR}, T) = 1.01 - 1.43V_{GR} + 0.165\left(\frac{T}{T_0}\right) - 16.8H_{GR}V_{GR} - 1.1H_{GR}V_{GR}\left(\frac{T}{T_0}\right) + 16H_{GR}^2V_{GR}^2 \quad (7b)$$

$$f_\rho(V_{GR}, T) = 1.01 - 2.01V_{GR}^2 - 0.0131\left(\frac{T}{T_0}\right) \quad (7c)$$

The GP algorithm introduces modification functions $[f_E, f_\nu, f_\rho](H_{GR}, V_{GR}, T)$ that enhance the conventional Halpin-Tsai micromechanics model and Rule of Mixtures by incorporating the effects of H atom coverage (H_{GR}), GOr volume fraction (V_{GR}) and temperature (T). It is important to mention that $T/T_0 = 1$ as the ambient conditions are considered herein.

Figs. 2 and 3 demonstrate that the Poisson's ratios and elastic moduli of graphene metamaterial plates are highly tuneable, enabling the structure to exhibit exceptional performance under dynamic conditions. From Figs. 2(a) and 3(a), it is determined that the $H_{GR}X$ - $W_{GR}X$ patterns provide the metamaterial plate with the highest stiffness along with a combination of positive and negative Poisson's ratio layers. As such, the distribution patterns have been utilised for the parametric studies involving H atom coverage and GOr content. According to Fig. 3(b), H coverage of $H_{GR} = 100\%$ exhibits the most prevalent NPR behaviour and depicts relatively weaker functionally graded layers compared to other GOr folding degrees (Fig. 2(b)). From Figs. 2(c) and 3(c), it is determined that increasing the GOr content provides the metamaterial composite with improved strength and prevalence of auxetic behaviour.

3. Theoretical formulation

3.1. Fluid mechanics

Consider the flow of a viscous fluid interacting with an FG-GOEAM plate. The normalised governing equations for the two-dimensional, laminar, unsteady, incompressible flow at $Re = 200$ and negligible dissipation effect can be expressed in the following forms:

$$\text{Continuity: } \frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \quad (8a)$$

$$\text{Momentum - } x : \frac{\partial U}{\partial t} + \frac{\partial UU}{\partial X} + \frac{\partial VU}{\partial Y} = -\frac{\partial P}{\partial X} + \frac{1}{Re} \left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2} \right) \quad (8b)$$

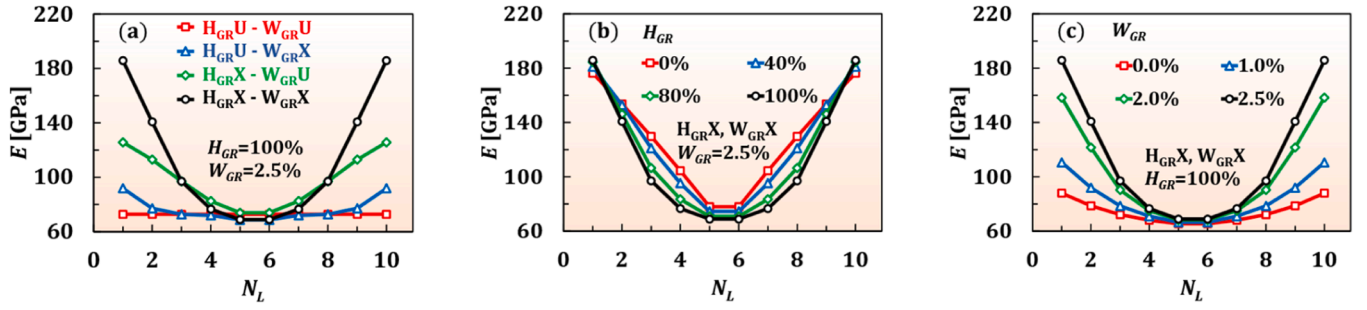


Fig. 2. Layer-wise distribution of Young's modulus (E) based on (a) GOrI and metamaterial distribution patterns, (b) H atom coverage, and (c) GOrI weight fraction.

$$\text{Momentum} - y : \frac{\partial V}{\partial t} + \frac{\partial UV}{\partial X} + \frac{\partial VV}{\partial Y} = -\frac{\partial P}{\partial Y} + \frac{1}{Re} \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) \quad (8c)$$

where Re is the Reynolds number based on the circular cylindrical dimension, and t is the dimensionless time. It should be noted that the flow conditions should be maintained at low Reynolds numbers in order for the Navier Stokes equations to remain pertinent. If this condition is not met, the aforementioned equations cannot remain applicable due to the presence of turbulence at higher Reynolds number. Based on various previous studies that proposed $Re \in [60, 200]$ [37,38], $Re = 200$ is used in this paper unless stated otherwise as it allows to isolate specific phenomena related to VIV and understand their underlying mechanisms before scaling up the analysis to more complex scenarios. The dimensionless variables are defined as below:

$$U = \frac{u}{U_\infty}, \quad V = \frac{v}{U_\infty}, \quad X = \frac{x}{D}, \quad Y = \frac{y}{D}, \quad P = \frac{p - p_\infty}{\rho U_\infty^2}, \quad t = \frac{U_\infty \tau}{D}, \quad Re = \frac{\rho U_\infty D}{\mu} \quad (9)$$

where u and v are the velocity components in the x - y directions, respectively, p is the pressure, U_∞ is the free stream velocity and τ is time. A parabolic velocity profile is prescribed at the left inlet

$$U_f = 1.5 U_\infty \frac{y(H_d - y)}{(H_d/2)^2}. \quad (10)$$

such that the mean inflow velocity is U_∞ and the maximum of the inflow velocity profile is $1.5 U_\infty$. The current analysis is transient, and defining the inlet velocity abruptly can lead to discontinuity in time and failure of

$$F_y = 1[N] \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}; \text{ where } x = 1.5, \quad \sigma = 1.5 \times 10^{-2} \quad (11)$$

where x is the location value for the Gaussian pulse and σ is the standard deviation. The dimensionless time t and maximum von Mises stress $\tilde{\sigma}_{max}$ is given by

$$t = \frac{\tau}{l_p} \sqrt{\frac{A_{110}}{I_{10}}}, \quad \tilde{\sigma}_{max} = \frac{\sigma_{max} w_p}{A_{110}} \quad (12)$$

where τ is the dimensional time, l_p is the length of the plate, σ_{max} is the dimensional maximum von Mises stress, w_p is the thickness of the plate, A_{110} and I_{10} are the extensional stiffness and inertia terms of the pure metallic plate under ambient conditions (ambient temperature, $T_0 = 300$ K).

3.2. Smoothing techniques

The mesh deformation can be determined using the nonlinear Yeoh smoothing type, offering the capability to represent a broad range of deformations. The current study employs the Yeoh smoothing method due to its applicability to small deformation problems [39]. In comparison to alternative methods [40–42], the application of Yeoh smoothing partially inhibits additional deformation in these areas and efficiently distributes mesh deformation uniformly throughout the domain. The strain energy W associated with Yeoh smoothing is expressed as follows:

$$W = \frac{1}{2} \int C_1(I_1 - 3) + C_2(I_1 - 3)^2 + C_3(I_1 - 3)^3 + \kappa(J - 1)^2 dV; \quad C_1 = 1, \quad C_3 = 0, \quad C_2 = 10 \quad (13)$$

the solution to converge. As such, the inlet velocity function is multiplied by a second-order step smoothing function. The Heaviside step function transitions from 0.0 to 1.0 s with the location of the start of the step at 0.5 s.

The outflow condition serves to specify a reference value for the pressure variable p . In the context of incompressible scenarios, this value can be arbitrarily defined. However, in the context of compressible structures, it significantly impacts stress and, consequently, the deformation of the solid. In this proposed approach, the reference pressure at the outflow is a zero-mean value.

A point load of 1 N as a Gaussian pulse is placed at the tip of the plate to avoid abrupt changes in the plate deformation that could lead to numerical instabilities or unrealistic responses in the simulation. Additionally, using a Gaussian pulse allows the control of the duration and shape of the excitation, making it easier to analyse the dynamic response of the system

where κ is an artificial bulk modulus while C_1 , C_2 , and C_3 are other artificial material properties. The value of Stiffening factor C_2 controls the nonlinear hardening of artificial materials under deformation, which is set to 10 in this simulation, and I_1 is Cauchy–Green deformation tensor. The Yeoh smoothing technique responds to element distortion by sharply increasing the stiffness of distorted elements. As a result, further distortion is prevented in these regions, thereby effectively spreading the mesh deformation more evenly over the domain away from moving boundaries. Yeoh smoothing produces comparatively better results to other smoothing techniques and allows the largest displacement of boundaries before mesh elements become inverted.

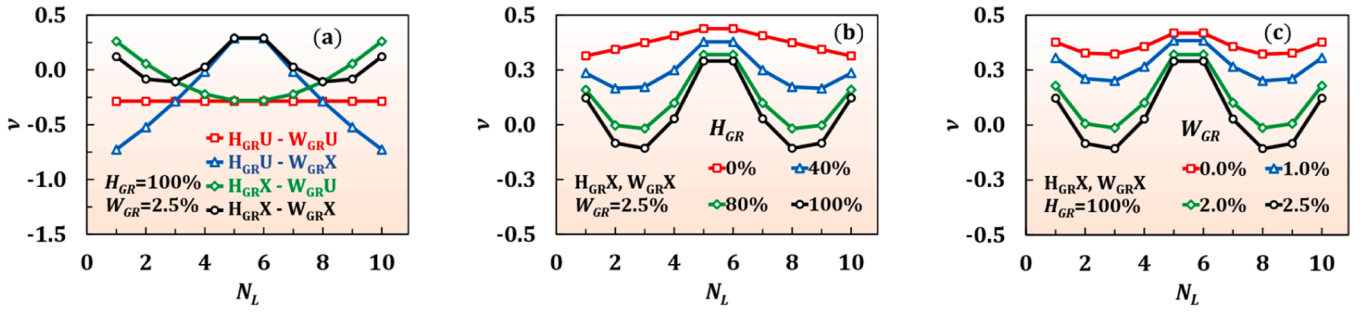


Fig. 3. Layer-wise distribution of Poisson's ratio (ν) based on (a) GORI and metamaterial distribution patterns, (b) H atom coverage, and (c) GORI weight fraction.

4. Results and discussion

4.1. Convergence study

The appropriate mesh grid and functionally graded layer numbers are determined via a comprehensive convergence test.

The mesh grid information under consideration has been tabulated (Table 1). It should be noted that defining the minimum and maximum mesh element sizes prevents the mesh grid from extreme fineness and coarseness, respectively. The computation time is observed to be 10–12 min. This outcome is achieved with the following computation resources: Processor: AMD Ryzen 7 3700X 8-Core Processor, 3600 MHz, 8 Cores, 16 Logical Processors. The convergence study results for the dimensionless maximum deflection at the plate tip (w) and dominant vibrating frequency (f) are presented in Table 2, wherein, it is observed that the computed results do not change much with improved refinement. Therefore, the conservative mesh distribution Mesh_3 and functionally graded layers $N_L = 10$ are selected for the following parametric study.

The mesh distribution in the computational domain is shown in Fig. 4. The whole domain consists of unstructured triangular mesh, with boundary layer mesh in the vicinity of the cylinder and near the surfaces of the plate.

4.2. Validation study

The validation of the finite element-based solver is presented in Fig. 5. Fig. 5(a) describes the variation of the linear fundamental frequencies, wherein, reductions are initially observed as the GORI weight fraction increases between 0.0 % and 1.0 %. Further increasing the GORI weight fraction by more than 1.5 %, improves the linear fundamental frequencies of the beam. The GORI/metallic nanocomposites transform into auxetic metamaterials, leading to the higher NPR and Young's modulus, and consequently enhanced beam stiffness. This observation is consistent with the material properties depicted in Figs. 2(c) and 3(c). Fig. 5(b) depicts the mean transverse displacement (W) from fluid-structure interaction tests where the flow around an elastic object result in periodic solutions. It shows the behaviour of the structure and

flow over time, with repeated oscillations occurring at regular intervals [35]. It can be seen that the present results are in good agreement with those reported in the literature, indicating the validity of the present model and solution method.

4.3. Parametric study

In this section, the study investigates the dynamic response of FG-GOEAM plates under the action of vortex-induced vibration. Unless stated otherwise, the following properties apply to the FG-GOEAM plate: cylinder diameter $D = 0.1$ m, metamaterial and GORI distribution $H_{GR}X$ -

Table 2

Convergence test of the numbers of mesh grid and functionally graded layer ($H_{GR}U$ and $W_{GR}U$) with respect to the dimensionless maximum deflection at the plate tip (w) and dominant vibrating frequency (f), where W and F are the dimensional maximum deflection and dominant frequency in Hz, respectively.

Mesh (N_L =10)	$w = W/w_p$	$f =$ $FL\sqrt{\frac{I_{10}}{A_{110}}}$	N_L (Mesh_3)	$w = W/w_p$	$f =$ $FL\sqrt{\frac{I_{10}}{A_{110}}}$
Mesh_1	1.76×10^{-5}	0.06208	8	1.76×10^{-5}	0.06208
Mesh_2	1.76×10^{-5}	0.06208	10	1.76×10^{-5}	0.06208
Mesh_3	1.76×10^{-5}	0.06208	12	1.76×10^{-5}	0.06208
Mesh_4	1.76×10^{-5}	0.06208	14	1.76×10^{-5}	0.06208
Mesh_5	1.76×10^{-5}	0.06208	16	1.76×10^{-5}	0.06208

Table 1

Mesh grid information.

Mesh	Mesh_1	Mesh_2	Mesh_3	Mesh_4	Mesh_5
Max mesh element size	1.16×10^{-1}	3.51×10^{-2}	1.86×10^{-2}	1.30×10^{-2}	7.02×10^{-3}
Min mesh element size	1.76×10^{-2}	7.02×10^{-4}	1.05×10^{-4}	4.39×10^{-5}	2.63×10^{-5}
Max element growth rate	2.00	1.40	1.30	1.25	1.20
Curvature factor	1.00	0.40	0.30	0.25	0.25
Number of elements	1261	1398	2565	2631	2797

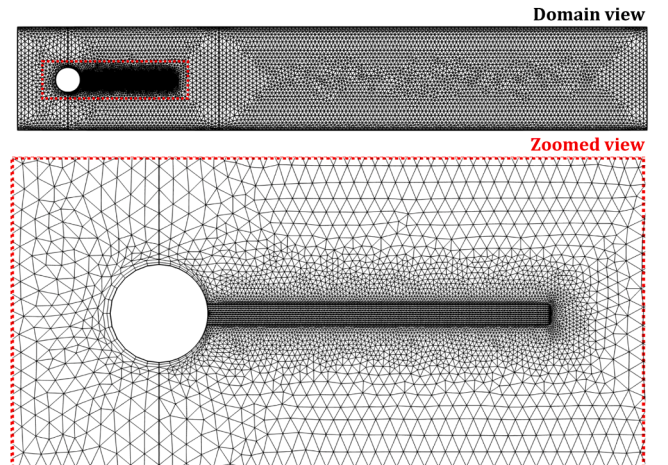


Fig. 4. Mesh distribution at the surfaces of the circular cylinder and FG-GOEAM plate (Mesh_3 and $N_L = 10$ layers).

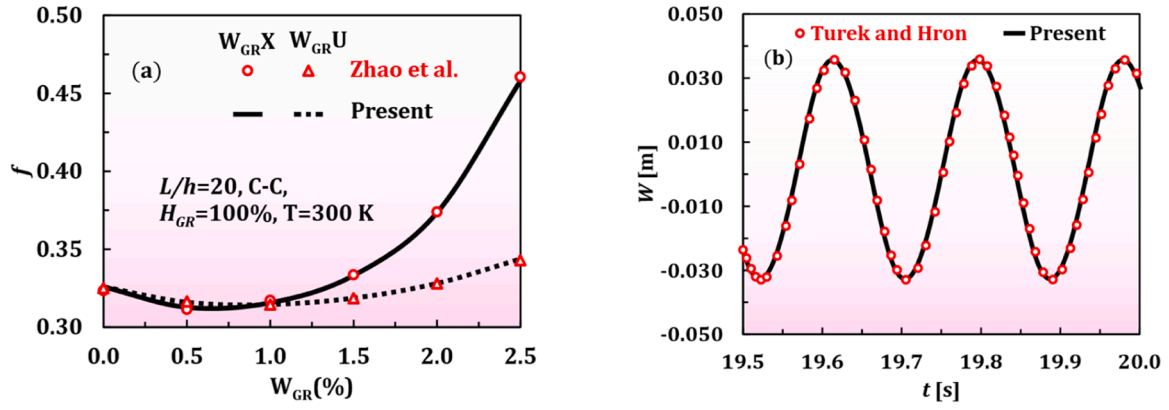


Fig. 5. (a) Comparison of the dimensionless fundamental frequencies obtained via the present methodology and that of Zhao, et al. [43]. (b) Comparison between the periodic solutions of the FSI tests conducted by Turek and Hron [35] and the current methodology.

Table 3

The relevant metallic copper matrix and graphene material properties at ambient conditions (300 K) where E , ρ , and ν are Young's modulus, density, and Poisson's ratio, respectively.

Material properties	Cu	Graphene
E	65.790 GPa	929.570 GPa
ρ	8.800 g/cm ³	1.800 g/cm ³
ν	0.387	0.220

W_{GRX} pattern, Gori weight fraction $W_{GR}=2.5\%$, $H_{GR}=100\%$ and step time $\Delta t = 5 \times 10^{-2}$. The laminar flow is characterised by low Reynolds numbers, ensuring the capacity to deform the plate structure. A typical fluid candidate for such experiments is glycerine with a fluid density of 1260 kg/m^3 , kinematic viscosity of $1127 \times 10^{-6} \text{ m}^2/\text{s}$ and dynamic viscosity of $1420 \times 10^{-3} \text{ kg/ms}$. As such Turek and Hron [35] recommend adopting the following benchmark settings for FSI problems: fluid density $\rho_f = 1000 \text{ kg/m}^3$, and fluid kinematic viscosity $\nu_f = 1000 \times 10^{-6} \text{ m}^2/\text{s}$.

For the following study, a metallic copper (Cu) matrix is selected due

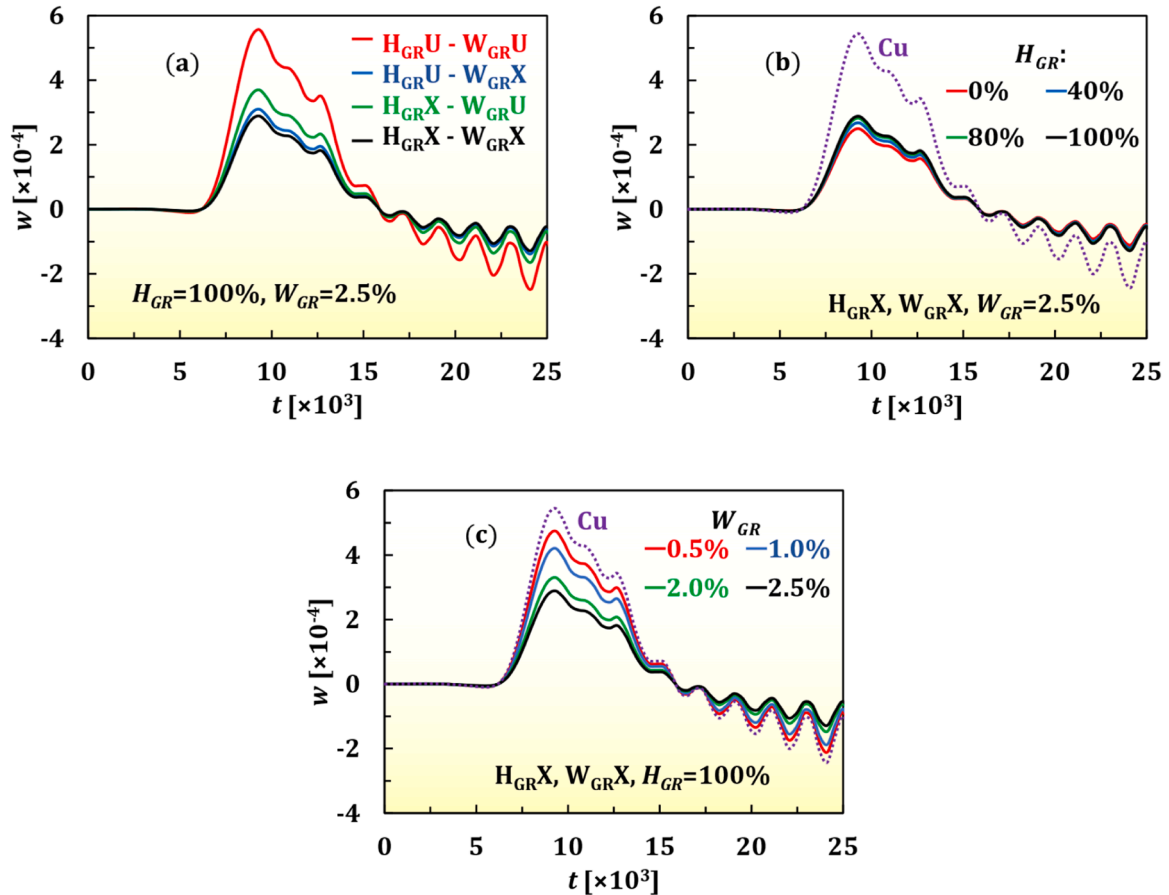


Fig. 6. Effect of the (a) metamaterial and Gori distributions, (b) hydrogen atom coverage and (c) Gori weight fractions on the time histories of the dimensionless transverse deflections at the plate tip of the FG-GOEAM plate.

to the favourable interfacial bonding with the GOr. The material properties of the Cu matrix and graphene reinforcements are provided in Table 3. The geometric length and thickness of the graphene are 83.76 Å and 3.4 Å, respectively.

Fig. 6 depicts the effect of various GOr properties on the time histories of the transverse plate tip displacements. From Fig. 6(a), the $H_{GR}U-W_{GR}U$ combination pattern exhibits the highest amplitudes compared to other metamaterial and GOr distribution patterns while the $H_{GR}X-W_{GR}X$ pattern exhibits the lowest deflection amplitude. For example, at $t = 9316.90$, the dimensionless transverse deflections are $w = 5.57 \times 10^{-4}$ and $w = 2.89 \times 10^{-4}$ for the $H_{GR}U-W_{GR}U$ and $H_{GR}X-W_{GR}X$ patterns, respectively. According to Fig. 2(a), the superior vibrational deflection performance of the $H_{GR}X-W_{GR}X$ pattern is attributed to the surface layers that possess high elastic moduli. Fig. 6(b) displays the gradual increase in the plate tip deflection in the transverse direction as the H atom coverage increases. Increasing the H coverage facilitates the transformation of the GOr/Cu composite to GOr metamaterial, thereby deteriorating the plate strength (Fig. 2(b) and Fig. 3(b)). Nevertheless, the GOr metamaterial plate with $H_{GR} = 100\%$ exhibits a 47 % lower tip deflection compared to the pristine Cu plate at $t = 9316.90$. Fig. 6(c) shows a decrease in the plate tip deflection as a result of increasing the graphene weight fraction within the structure. As observed in Fig. 3(c), the addition of GOr within the structure improves its stiffness thereby enhancing its deformation resistance.

Fig. 7 displays the effect of various GOr properties on the time histories of the maximum von Mises stresses. As observed from Fig. 7(a), It is observed that the $H_{GR}X-W_{GR}X$ and $H_{GR}U-W_{GR}X$ exhibit $\bar{\sigma}_{max} = 4.57 \times 10^{-6}$ and $\bar{\sigma}_{max} = 8.40 \times 10^{-6}$ at $t = 9316.90$, respectively. $H_{GR}U-W_{GR}X$ experiences the highest maximum von Mises stress due to the relatively weaker surface layers that exhibit high NPR values (Fig. 3(a)). Weaker

layers have a lower loading carrying capacity. Fig. 7(b) and (c) show that the effects of the H coverage and GOr content on the maximum von Mises stress are minimal, however, a gradual increase in $\bar{\sigma}_{max}$ is observed as a result of increasing the H coverage and graphene content. From Fig. 2(b), the trend behaviours of the GOr folding degrees are quite similar, therefore, similar maximum von Mises stresses are observed. From Fig. 2(c), as the GOr content increases, the elastic modulus at the surface layers increases, thereby, increasing the loading carrying capacity of the plate structure.

Fig. 8 depicts the Kármán vortex streets downstream of the cylinder-plate assembly. The distinct wake patterns observed in Fig. 8 for the cylinder-plate assembly can be attributed to the influence of plate length on fluid mechanics. Shorter plates act as disruptive obstacles, intensifying vortex shedding due to abrupt flow disruptions and increased pressure differentials. In contrast, longer plates facilitate a more gradual interaction with the flow, leading to effective flow re-energisation along the extended surface. This re-energisation mitigates vortex shedding, thus, reducing its prevalence in the wake. The transition from shorter to longer plates signifies a shift in the dominant flow regime, highlighting the delicate balance between flow confinement and fluid re-energisation as crucial factors in determining wake patterns.

From Fig. 9(a), it is observed that the natural frequencies of both the metamaterial and pure Cu plates are well above the vortex shedding frequency 4.4 Hz. Nevertheless, reinforcing the metallic Cu plate with graphene origami further elevates the natural frequency of the composite structure. Therefore, these structures operate within a regime where vortex shedding is unlikely to be a dominant phenomenon. This augmentation may be attributed to the unique mechanical properties of graphene, which contribute to enhanced stiffness and structural integrity (Fig. 2). As a result, the composite structure exhibits a heightened

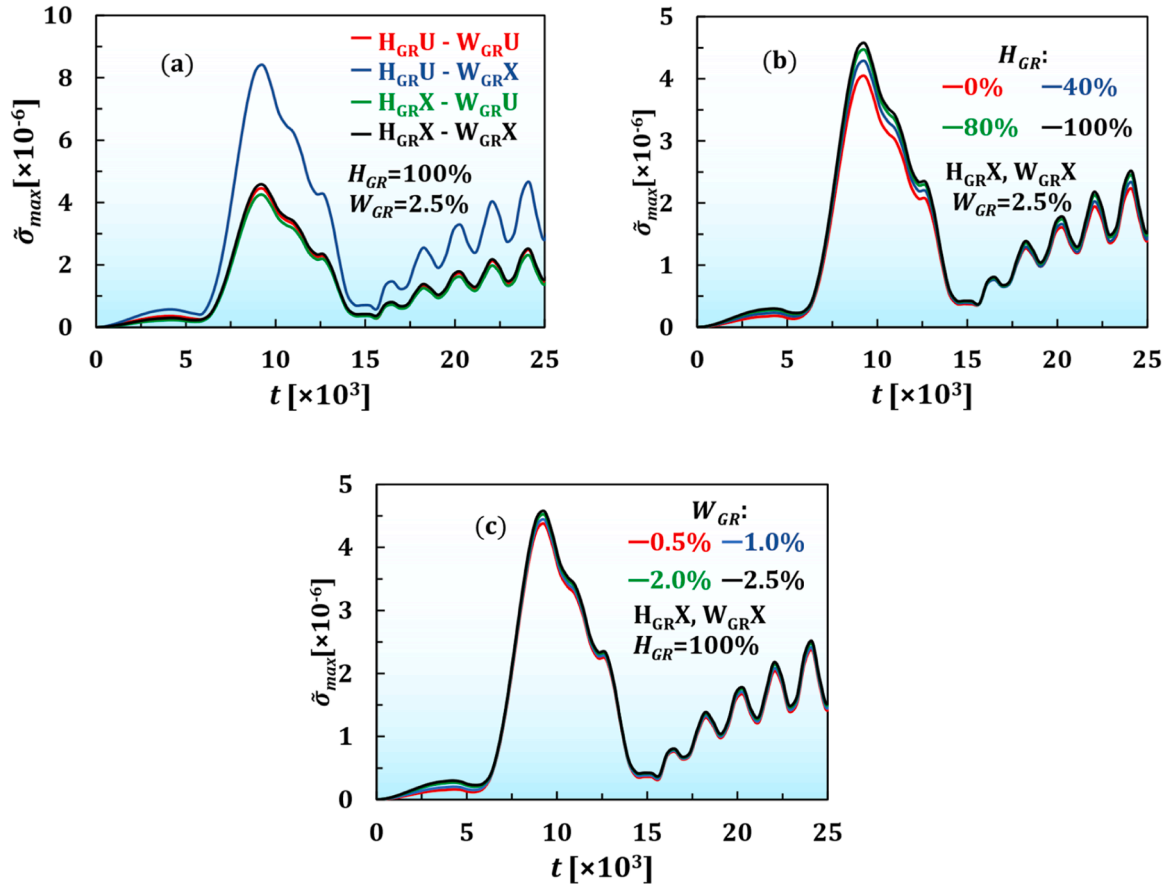


Fig. 7. Effect of the (a) GOr and metamaterial distributions, (b) hydrogen atom coverage and (c) GOr weight fractions on the time histories of the dimensionless maximum von Mises stress of the FG-GOEAM plate.

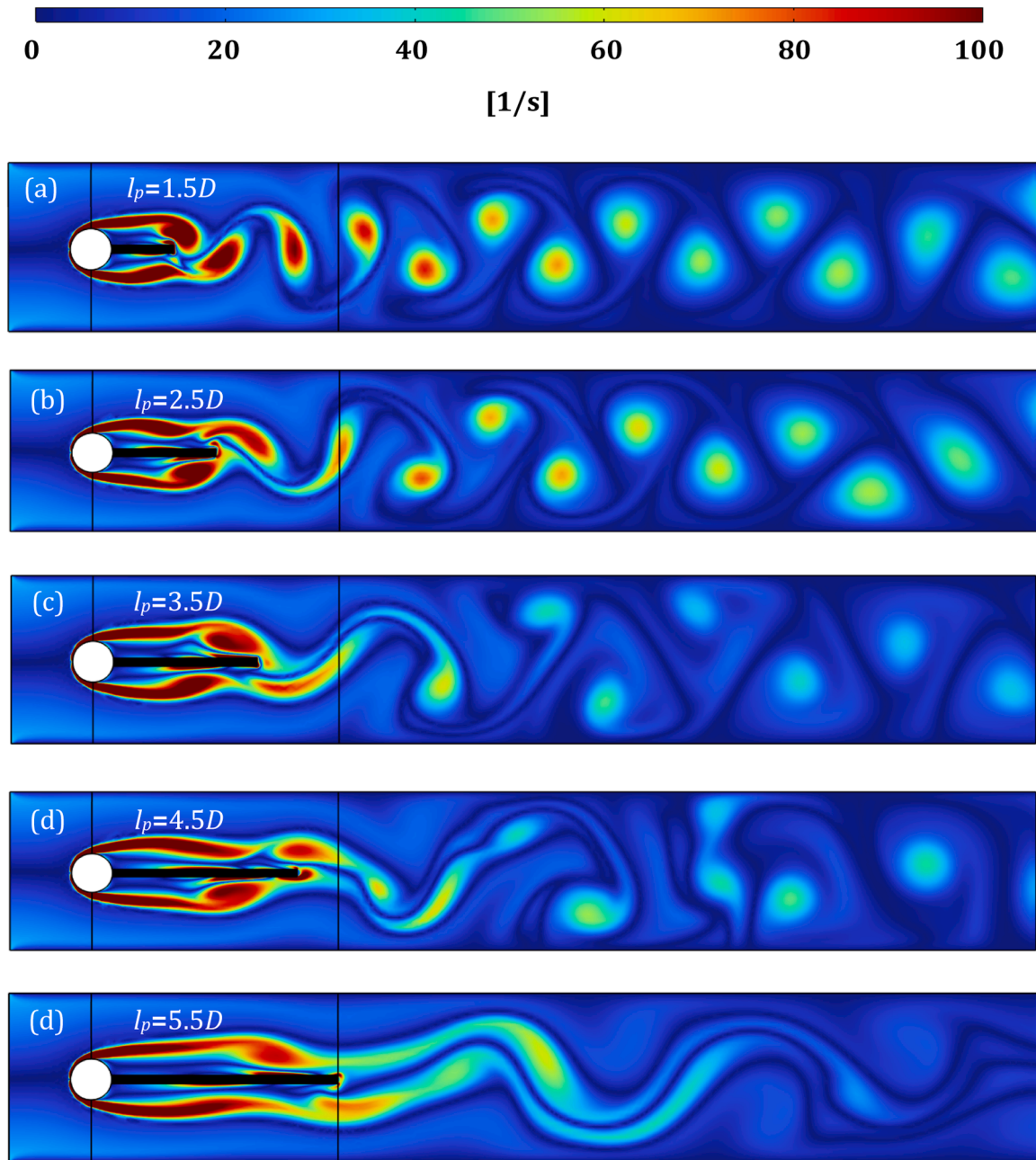


Fig. 8. Instantaneous vorticity magnitude contour plots and wake patterns of the cylinder-plate assembly at various plate lengths ($D = 0.1$ m).

resilience to dynamic forces, manifesting in an increased natural frequency. Fig. 9(b) illustrates the inverse relationship between the circular cylinder (D) and vortex shedding frequency (f_d). Larger cylinder diameters correspond to lower Strouhal numbers, indicative of a more stable and predictable vortex-shedding behaviour, at constant vortex-shedding frequencies and inlet velocities (U). From Fig. 9(c), it is observed that higher Reynolds numbers cause larger deformations because they correspond to higher flow velocities. As the flow velocity increases, it exerts a stronger force on the cylinder, leading to larger deformations in its wake. This is due to increased momentum in the fluid flow, which results in greater pressure fluctuations and vortices around the cylinder. Fig. 9(d) depicts the influence of the slenderness ratio on the plate deflection behaviour. The maximum deflection is $w = 3.6 \times 10^{-3}$ when the slenderness ratio $l_p/w_p = 35$ while the deflection becomes $w = 5.45 \times 10^{-4}$ when the slenderness ratio $l_p/w_p = 18$. It can be observed that increasing the slenderness ratio results in more pronounced

deflection due to the reduced stiffness of the plate.

5. Conclusions

In the present study, the effects of VIV on the dynamic response of FG-GOEAM splitter plates are investigated. The tip displacements and von Mises stresses as a result of the graphene properties and fluid conditions are considered. The analyses are accomplished numerically by considering a two-dimensional unsteady fluid flow ($Re = 200$) over a circular cylinder-plate assembly. The findings are summarised below:

- (1). The $H_{GR}X-W_{GR}X$ pattern exhibits the lowest deflection amplitudes, attributed to the surface layers with high elastic moduli, while the $H_{GR}U-W_{GR}U$ combination pattern shows the highest deflection amplitudes. The incorporation of GOr within the structure improves stiffness and enhances deformation

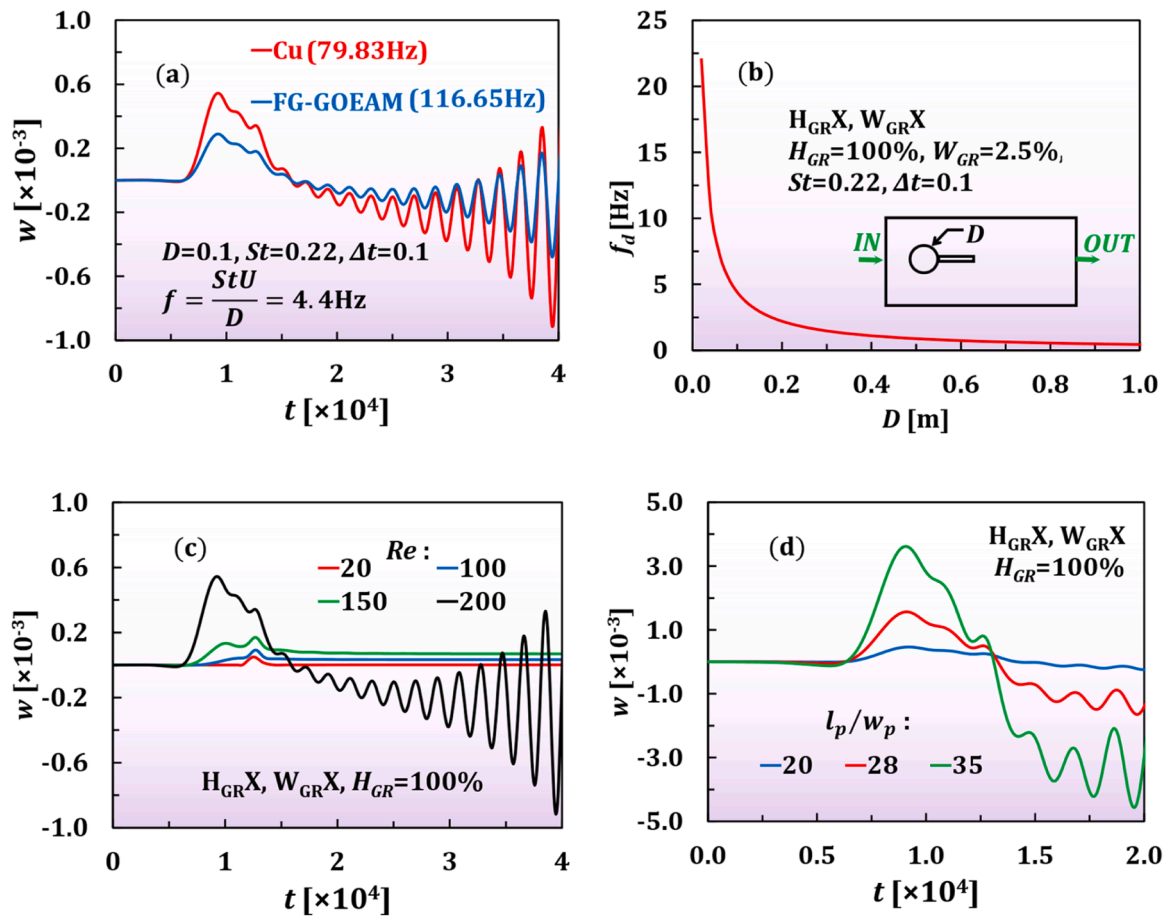


Fig. 9. (a) Transverse displacements and dominant frequencies of FG-GOEAM and pure Cu plates, (b) Relationship between the cylinder diameter in m (D) and the vortex-shedding frequency in Hz (f_d), (c) Effect of the Reynolds number on the dimensionless transverse displacements FG-GOEAM plates, (d) Effect of slenderness ratio on the dimensionless transverse displacements FG-GOEAM plates.

resistance, as evidenced by the observed decrease in plate tip deflection with increasing graphene weight fraction.

- (2). Higher hydrogen coverage facilitates the transformation of the GOr/Cu composite to GOr metamaterial, leading to a deterioration in plate strength. However, even with $H_{GR} = 100\%$, the GOr metamaterial plate exhibits a 47 % lower tip deflection compared to the pristine Cu plate, highlighting the superior performance of the metamaterial in resisting deformation.
- (3). $H_{GR}U-W_{GR}X$ experiences the highest maximum von Mises stress due to weaker GOr layers, while increasing hydrogen atom coverage and GOr weight fraction result in a gradual increase in stress. However, the effects of hydrogen coverage and GOr on the maximum von Mises stress are minimal, emphasizing the dominance of distribution patterns in influencing stress levels.
- (4). The investigation into the flow dynamics around the cylinder-plate assembly reveals that shorter plates act as obstructions, enhancing vortex intensity, while longer plates re-energize the flow, reducing vortex prevalence.
- (5). The dominant frequencies of both FG-GOEAM and pure Cu plates are found to be well above the vortex shedding frequency, with the incorporation of graphene origami further elevating the dominant frequency. Higher Reynolds numbers lead to increased deformations in FG-GOEAM plates due to higher flow velocities, resulting in stronger forces and larger deformations in the wake of the cylinder.

CRediT authorship contribution statement

Bill Murari: Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Shaoyu Zhao:** Writing – review & editing, Methodology, Formal analysis. **Yihe Zhang:** Writing – review & editing, Supervision, Resources. **Jie Yang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research received complete financial support from the Australian Research Council grant under the Discovery Project scheme (DP210103656, DP230100548). The authors express their deep gratitude for this funding.

References

- [1] S. Babaee, J. Shim, J.C. Weaver, E.R. Chen, N. Patel, K. Bertoldi, 3D soft metamaterials with negative Poisson's ratio, *Adv. Mater.* 25 (36) (2013) 5044–5049, <https://doi.org/10.1002/adma.201301986>.
- [2] L. Mizzi, K.M. Azzopardi, D. Attard, J.N. Grima, R. Gatt, Auxetic metamaterials exhibiting giant negative Poisson's ratios, *physica status solidi (RRL)–Rapid Res. Lett.* 9 (7) (2015) 425–430, <https://doi.org/10.1002/pssr.201510178>.
- [3] H. Yasuda, J. Yang, Reentrant origami-based metamaterials with negative Poisson's ratio and bistability, *Phys. Rev. Lett.* 114 (18) (2015) 185502, <https://doi.org/10.1103/PhysRevLett.114.185502>.
- [4] S. Zhao, Y. Zhang, Y. Zhang, W. Zhang, J. Yang, S. Kitipornchai, Genetic programming-assisted micromechanical models of graphene origami-enabled metal metamaterials, *Acta Mater.* 228 (2022) 117791, <https://doi.org/10.1016/j.actamat.2022.117791>, /04/15/2022.
- [5] F. Meng, et al., Negative Poisson's ratio in graphene Miura origami, *Mech. Mater.* 155 (2021) 103774, <https://doi.org/10.1016/j.mechmat.2021.103774>, 2021/04/01/.
- [6] N. Wei, Y. Chen, Y. Zhang, J.-C. Zheng, J. Zhao, Y.-W. Mai, Crease-induced targeted cutting and folding of graphene origami, *Carbon N Y* 165 (2020) 259–266, <https://doi.org/10.1016/j.carbon.2020.04.058>, 2020/09/15/.
- [7] D.T. Ho, H.S. Park, S.Y. Kim, U. Schwingenschlögl, Graphene origami with highly tunable coefficient of thermal expansion, *ACS Nano* 14 (7) (2020) 8969–8974, <https://doi.org/10.1021/acsnano.0c03791>, 2020/07/28.
- [8] J. Cai, E. Estakhrianhaghghi, A. Akbarzadeh, Functionalized graphene origami metamaterials with tunable thermal conductivity, *Carbon N Y* 191 (2022) 610–624, <https://doi.org/10.1016/j.carbon.2022.02.008>, 2022/05/01/.
- [9] S. Zhao, Y. Zhang, Y. Zhang, J. Yang, S. Kitipornchai, Graphene origami-enabled auxetic metallic metamaterials: an atomistic insight, *Int. J. Mech. Sci.* 212 (2021) 106814, <https://doi.org/10.1016/j.ijmecsci.2021.106814>, 2021/12/15/.
- [10] B. Saleh, et al., 30 Years of functionally graded materials: an overview of manufacturing methods, Applications and Future Challenges, *Compos. Part B* 201 (2020) 108376, <https://doi.org/10.1016/j.compositesb.2020.108376>, 2020/11/15/.
- [11] Y. Miyamoto, W. Kaysser, B. Rabin, A. Kawasaki, R.G. Ford, *Functionally Graded Materials: Design, Processing and Applications*, Springer Science & Business Media, 2013.
- [12] G. Udupa, S.S. Rao, K. Gangadharan, Functionally graded composite materials: an overview, *Procedia Mater. Sci.* 5 (2014) 1291–1299, <https://doi.org/10.1016/j.mspro.2014.07.442>.
- [13] P. Zhao, S. Wang, S. Guo, Y. Chen, Y. Ling, J. Li, Bonding W and W–Cu composite with an amorphous W–Fe coated copper foil through hot pressing method, *Mater. Des.* 42 (2012) 21–24, <https://doi.org/10.1016/j.matdes.2012.05.057>.
- [14] S. Kumar, K.M. Reddy, A. Kumar, G.R. Devi, Development and characterization of polymer–ceramic continuous fiber reinforced functionally graded composites for aerospace application, *Aerosp. Sci. Technol.* 26 (1) (2013) 185–191, <https://doi.org/10.1016/j.ast.2012.04.002>.
- [15] S. Ram, K. Chattopadhyay, I. Chakrabarty, High temperature tensile properties of centrifugally cast in-situ Al–Mg2Si functionally graded composites for automotive cylinder block liners, *J. Alloys Compd.* 724 (2017) 84–97, <https://doi.org/10.1016/j.jallcom.2017.06.306>.
- [16] G. Anné, K. Vanmeensel, J. Vleugels, O. Van der Biest, Electrophoretic deposition as a novel near net shaping technique for functionally graded biomaterials, in: *Materials Science Forum*, 492, Trans Tech Publ, 2005, pp. 213–218, <https://doi.org/10.4028/www.scientific.net/MSF.492-493.213>.
- [17] X. Miao, D. Sun, Graded/gradient porous biomaterials, *Materials (Basel)* 3 (1) (2009) 26–47, <https://doi.org/10.3390/ma3010026>.
- [18] W. Pompe, et al., Functionally graded materials for biomedical applications, *Mater. Sci. Eng.* 362 (1–2) (2003) 40–60, [https://doi.org/10.1016/S0921-5093\(03\)00580-X](https://doi.org/10.1016/S0921-5093(03)00580-X).
- [19] M.-J. Suk, S.-I. Choi, J.-S. Kim, Y.D. Kim, Y.-S. Kwon, Fabrication of a porous material with a porosity gradient by a pulsed electric current sintering process, *Met. Mater. Int.* 9 (2003) 599–603, <https://doi.org/10.1007/BF03027261>.
- [20] Y. Lv, J. Zhang, J. Wu, L. Li, Mechanical and thermal postbuckling of functionally graded graphene origami-enabled auxetic metamaterials plates, *Eng. Struct.* 298 (2024) 117043, <https://doi.org/10.1016/j.engstruct.2023.117043>, 2024/01/01/.
- [21] F. Chen, X. Qiu, K.A. Alnowibet, Size-dependent nonlinear vibrations of functionally graded origami-enabled auxetic metamaterial plate: application of artificial intelligence networks for solving the engineering problem, *Mater. Today Commun.* 38 (2024) 108232, <https://doi.org/10.1016/j.mtcomm.2024.108232>, 2024/03/01/.
- [22] B. Murari, S. Zhao, Y. Zhang, J. Yang, Graphene origami-enabled auxetic metamaterial tapered beams in fluid: nonlinear vibration and postbuckling analyses via physics-embedded machine learning model, *Appl. Math. Model.* 122 (2023) 598–613, <https://doi.org/10.1016/j.apm.2023.06.023>, 2023/10/01/.
- [23] B. Murari, S. Zhao, Y. Zhang, J. Yang, Static and dynamic instability of functionally graded graphene origami-enabled auxetic metamaterial beams with variable thickness in fluid, *Ocean Eng.* 280 (2023) 114859, <https://doi.org/10.1016/j.oceaneng.2023.114859>, 2023/07/15/.
- [24] J. Su, W. He, K. Zhang, Q. Zhang, Y. Qu, Vibration analysis of functionally graded porous cylindrical shells filled with dense fluid using an energy method, *Appl. Math. Model.* 108 (2022) 167–188, <https://doi.org/10.1016/j.apm.2022.03.028>, 2022/08/01/.
- [25] Q.-H. Pham, P.-C. Nguyen, V.K. Tran, T. Nguyen-Thoi, Isogeometric analysis for free vibration of bidirectional functionally graded plates in the fluid medium, *Defence Technol.* 18 (8) (2022) 1311–1329, <https://doi.org/10.1016/j.dt.2021.09.006>, 2022/08/01/.
- [26] Y. Hu, B. Yang, X. Chen, X. Wang, J. Liu, Modeling and experimental study of a piezoelectric energy harvester from vortex shedding-induced vibration, *Energy Convers. Manage.* 162 (2018) 145–158, <https://doi.org/10.1016/j.enconman.2018.02.026>.
- [27] S. Shukla, R. Govardhan, J. Arakeri, Dynamics of a flexible splitter plate in the wake of a circular cylinder, *J. Fluids Struct.* 41 (2013) 127–134, <https://doi.org/10.1016/j.jfluidstructs.2013.03.002>.
- [28] X. Sun, C.S. Suh, C. Sun, B. Yu, Vortex-induced vibration of a flexible splitter plate attached to a square cylinder in laminar flow, *J. Fluids Struct.* 101 (2021) 103206, <https://doi.org/10.1016/j.jfluidstructs.2020.103206>, 2021/02/01/.
- [29] Z. Zhou, S. Huang, Y. Wang, Numerical simulation study of the fluid-structure coupling effects of Kármán vortex street on airfoil motion with varying thickness, *Ocean Eng.* 286 (2023) 115459, <https://doi.org/10.1016/j.oceaneng.2023.115459>, 2023/10/15/.
- [30] P. Dey, Enhancement of thermo-fluid performance of square cylinder by dual splitter plates, *Int. J. Mech. Sci.* 238 (2023) 107849, <https://doi.org/10.1016/j.ijmecsci.2022.107849>.
- [31] M. Zhao, A review of recent studies on the control of vortex-induced vibration of circular cylinders, *Ocean Eng.* 285 (2023) 115389, <https://doi.org/10.1016/j.oceaneng.2023.115389>, 2023/10/01/.
- [32] W. Chen, C. Ji, D. Xu, Vortex-induced vibrations of two side-by-side circular cylinders with two degrees of freedom in laminar cross-flow, *Comput. Fluids* 193 (2019) 104288, <https://doi.org/10.1016/j.compfluid.2019.104288>, 2019/10/30/.
- [33] M. Ostolaza, J.I. Arrizubieta, A. Lamikiz, S. Plaza, N. Ortega, Latest developments to manufacture metal matrix composites and functionally graded materials through AM: a state-of-the-art review, *Materials (Basel)* 16 (4) (2023) 1746, <https://doi.org/10.3390/ma16041746>.
- [34] D. Gayen, R. Tiwari, D. Chakraborty, Static and dynamic analyses of cracked functionally graded structural components: a review, *Compos. Part B* 173 (2019) 106982, <https://doi.org/10.1016/j.compositesb.2019.106982>, 2019/09/15/.
- [35] S. Turek, J. Hron, Proposal for numerical benchmarking of fluid-structure interaction between an elastic object and laminar incompressible flow, in: H.-J. Bungartz, M. Schäfer (Eds.), *Fluid-Structure Interaction*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2006, pp. 371–385, https://doi.org/10.1007/3-540-34596-5_15, Eds.2006/.
- [36] S. Kaneko, T. Nakamura, F. Inada, M. Kato, 2 - Vibration Induced by Cross-Flow, in: S. Kaneko, T. Nakamura, F. Inada, M. Kato (Eds.), *Flow Induced Vibrations*, Elsevier, Amsterdam, 2008, pp. 29–106, Eds.
- [37] M. Zhao, L. Cheng, L. Lu, Vortex induced vibrations of a rotating circular cylinder at low Reynolds number, *Phys. Fluids* 26 (7) (2014), <https://doi.org/10.1063/1.4886196>.
- [38] S. Mittal, T.K. Prasanth, Vortex-induced vibrations of a circular cylinder at low Reynolds numbers, *J. Fluid Mech.* 594 (2008) 463–491, <https://doi.org/10.1017/S002212007009202>.
- [39] S. Ge-Zhang, X. Chen, H. Zhu, Y. Song, Y. Ding, J. Cui, Computer simulation of polyethylene terephthalate carbonated beverage bottle bottom structure based on manual-automatic double-adjustment optimization, *Polymers (Basel)* 14 (14) (2022) 2845, <https://doi.org/10.3390/polym14142845>.
- [40] L. Huang, H. Li, K. Zheng, K. Tian, B. Wang, Shape optimization method for axisymmetric disks based on mesh deformation and smoothing approaches, *Mech. Adv. Mater. Struct.* 30 (12) (2023) 2532–2555, <https://doi.org/10.1080/15376494.2022.2058658>.
- [41] A. B. Vante and T. C. Kanish, Fluid-structure interaction and experimental studies of passive check valve based piezoelectric micropump for biomedical applications, *Advances in Materials and Processing Technologies*, pp. 1–27, <https://doi.org/10.1080/2374068X.2023.2206176>.
- [42] I. El Jirari, A. El Baroudi, A. Ammar, Predictive dynamical behavior of liquid-filled microparticles partitioning in the vicinity of a coronary bifurcation under pulsatile blood flow and arterial distensibility: a two-dimensional study, *J. Fluids Struct.* 120 (2023) 103893, <https://doi.org/10.1016/j.jfluidstructs.2023.103893>, 2023/07/01/.
- [43] S. Zhao, Y. Zhang, Y. Zhang, J. Yang, S. Kitipornchai, A functionally graded auxetic metamaterial beam with tunable nonlinear free vibration characteristics via graphene origami, *Thin-Walled Struct.* 181 (2022) 109997, <https://doi.org/10.1016/j.tws.2022.109997>, 2022/12/01/.